

Topic Modeling

Simone Rebora

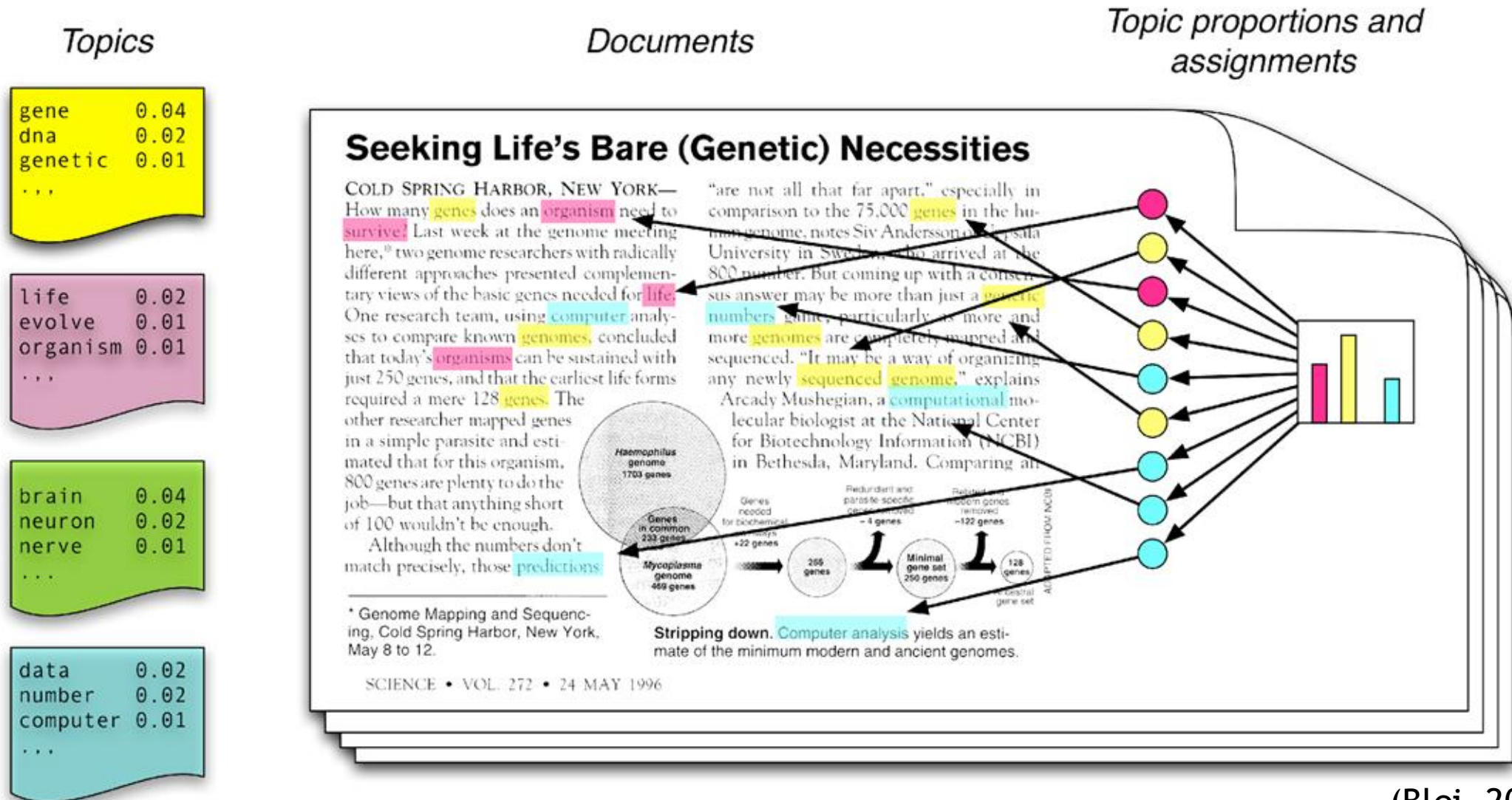
Topic Models

According to David Blei:

“Topic models are a suite of algorithms that **uncover the hidden thematic structure** in document collections. These algorithms help us develop new ways **to search, browse and summarize** large archives of texts”

(<http://www.cs.columbia.edu/~blei/topicmodeling.html>)

Topic Models



Topic Modeling Techniques

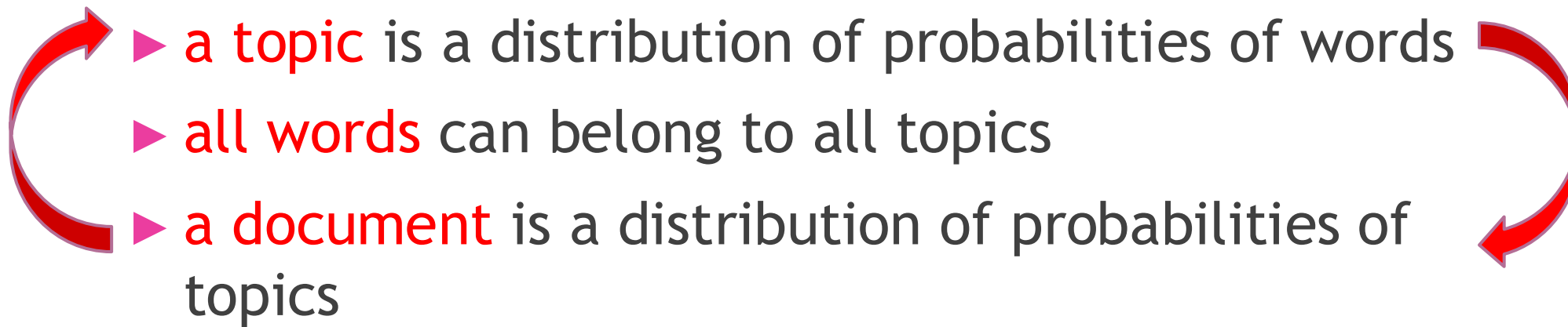
Older: Reinert method (1983), available in the Iramuteq software (<http://www.iramuteq.org/>)

Classic: Latent Dirichlet Allocation

Most recent: BERTopic, using BERT word embeddings (<https://maartengr.github.io/BERTopic/index.html>)

LDA Topic Models

LDA = Latent Dirichlet Allocation

- 
- ▶ a topic is a distribution of probabilities of words
 - ▶ all words can belong to all topics
 - ▶ a document is a distribution of probabilities of topics

LDA Topic Models

a topic:

sole (10.1%)
cuore (6.4%)
amore (4.7%)
...

a word:

amore

4.7%

7.1%

5.8%

*bad
poetry*

sentiments

12.4%

5.2%

15.8%

...

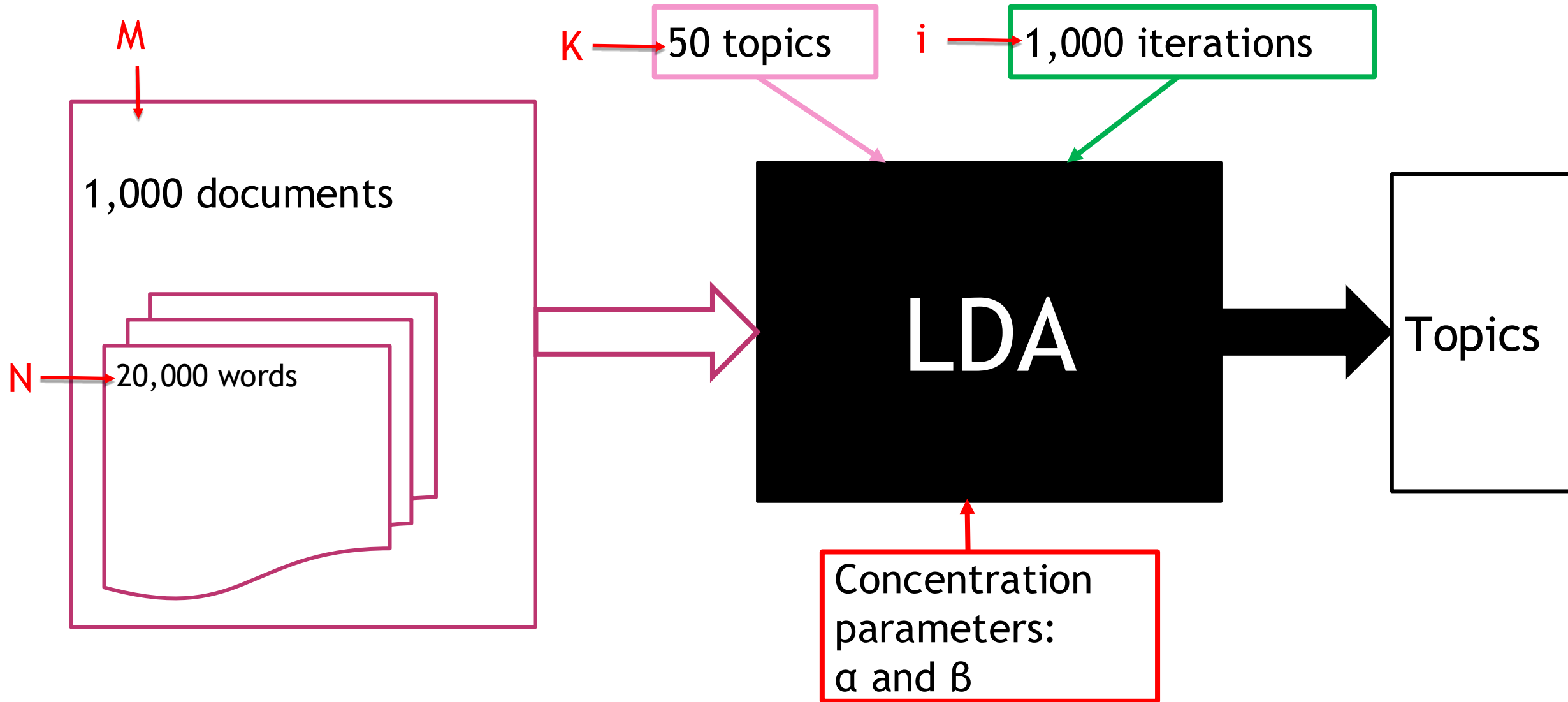
a document:



LDA: How Does it Work?

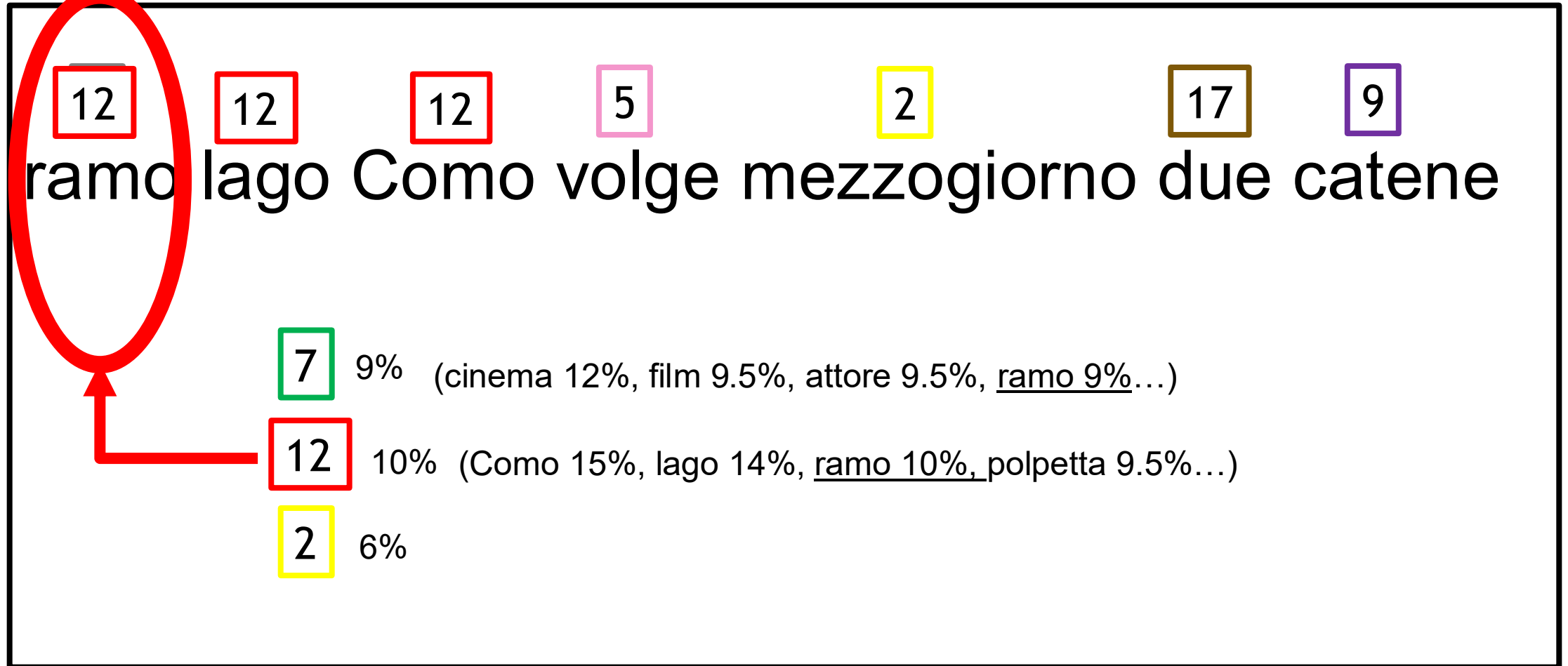
- ▶ Initialize topic assignments **randomly**
- ▶ **For each token** in each document:
 - ▶ **re-sample topic** for token,
given all other tokens and their current topic assignments
- ▶ Iterate n times!

LDA: How Does it Work?



LDA's «Black Box»

iteration #1,456



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Topic Modeling Genre: An Exploration of French Classical and Enlightenment Drama

[Christof Schöch](#) <christof_dot_schoech_at_uni-wuerzburg_dot_de>, University of Würzburg, Germany

Abstract

The concept of literary genre is a highly complex one: not only are different genres frequently defined on several, but not necessarily the same levels of description, but consideration of genres as cognitive, social, or scholarly constructs with a rich history further complicate the matter. This contribution focuses on thematic aspects of genre with a quantitative approach, namely Topic Modeling. Topic Modeling has proven to be useful to discover thematic patterns and trends in large collections of texts, with a view to class or browse them on the basis of their dominant themes. It has rarely if ever, however, been applied to collections of dramatic texts.

In this contribution, Topic Modeling is used to analyze a collection of French Drama of the Classical Age and the Enlightenment. The general aim of this contribution is to discover what semantic types of topics are found in this collection, whether different dramatic subgenres have distinctive dominant topics and plot-related topic patterns, and inversely, to what extent clustering methods based on topic scores per play produce groupings of texts which agree with more conventional genre distinctions. This contribution shows that interesting topic patterns can be detected which provide new insights into the thematic, subgenre-related structure of French drama as well as into the history of French drama of the Classical Age and the Enlightenment.

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“The data used in this study comes from the Théâtre classique collection maintained by Paul Fièvre (2007-2015). At the time of writing, this continually-growing, freely available **collection of French dramatic texts contained 890 plays published between 1610 and 1810,** thus covering the Classical Age and the Enlightenment.”
(Schöch, 2017)

topic 32 (1/60)

oser aimer coeur amour gloire

seul âme peine craindre doux secret vain offrir laisser hymen souffrir espoir connaître

croire plaire aveu prix princesse plaindre voeu

madame gloire choix flamme

topic 3 (6/60)

craindre secret connaître

être même expliquer attendre surprendre chercher esprit

trouver effet passer aimer seul crois entendre moins penser besoin

mystère ignorer doute soupçon temps part

soin oser paraître taire tenir croire

ami avoir cacher apprendre découvrir avouer sembler

topic 30 (53/60)

vers auteur

poète seul sujet génie muse mauvais

goût art talent

bon théâtre jouer rôle scène

temps ouvrage acteur nom merveille

nouveau trouver écrire rime premier public

esprit beau lire pièce

écrit prose commencer nommer sonnet plaisir

topic 34 (57/60)

monsieur malade

remède chose vif malade savant

science fou statue connaître

astrologie effet jours santé maladie corps folie

art charlatans avis cause

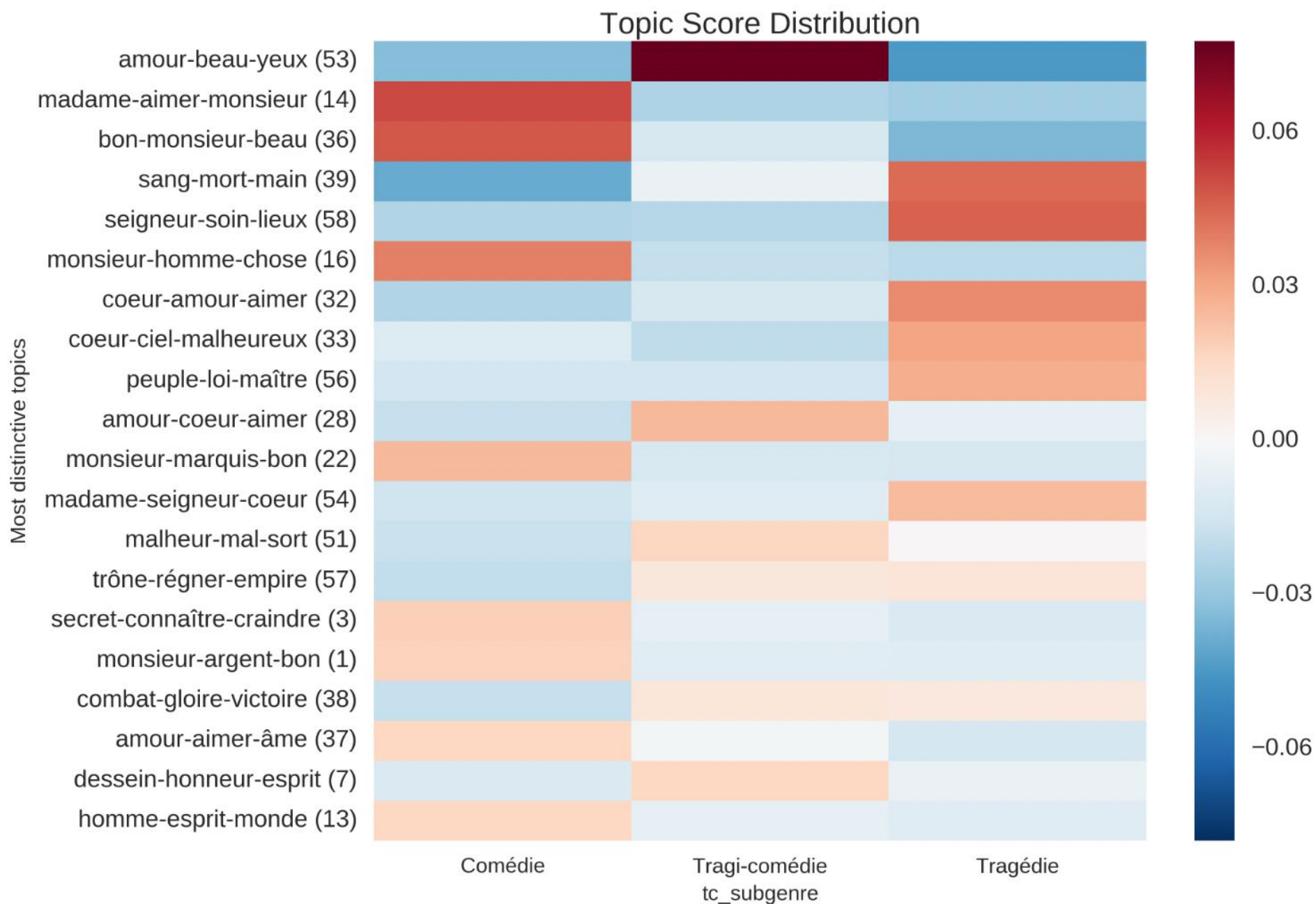
vapeur docteur humeur

médecin guérir

vrai goutte secours soutenir raison habile mourir

guérison médecine homme

(Schöch,
2017)



(Schöch,
2017)

TM Applications in Literary Studies

The image is a screenshot of the Journal of Digital Humanities (JDH) website. The header features the JDH logo and the journal's name. A navigation bar includes links for 'About', 'Volumes', and 'Submissions', along with an RSS feed subscription link. The main content area displays the 'Table of Contents for Vol. 2, No. 1 Winter 2012'. The table of contents lists sections: 'Introductions', 'Beginnings', 'Applications and Critiques', and two entries for 'Topic Modeling and Figurative Language' by Lisa M. Rhody. The article 'What Can Topic Models of PMLA Teach Us About the History of Literary Scholarship?' by Ted Underwood and Andrew Goldstone is highlighted. The article's abstract is visible, discussing the history of criticism and disciplinary history.

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Table of Contents for Vol. 2, No. 1 Winter 2012

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Beginnings

Applications and Critiques

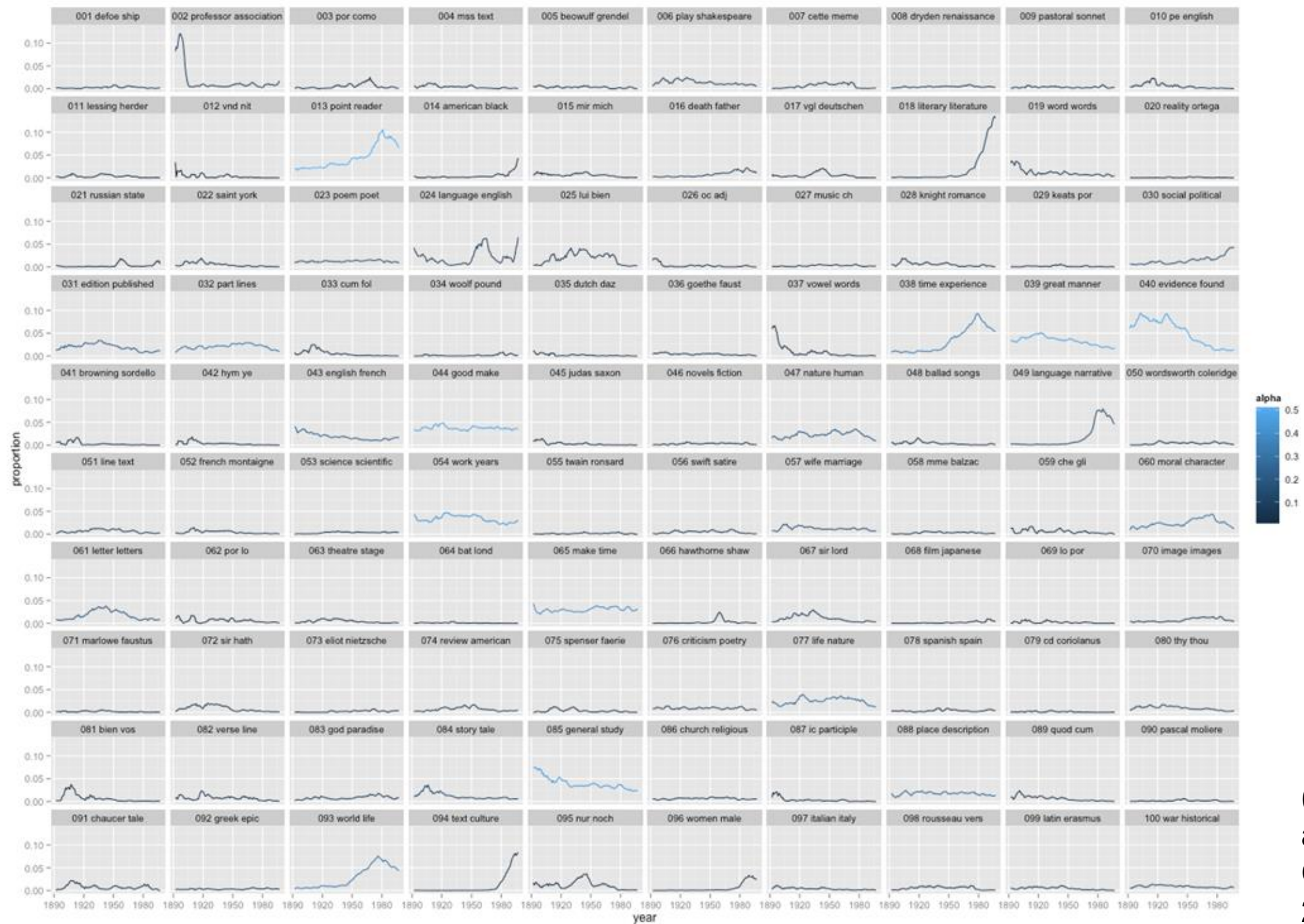
Topic Modeling and Figurative Language
Lisa M. Rhody

Topic Model Data for Topic Modeling and Figurative Language
Lisa M. Rhody

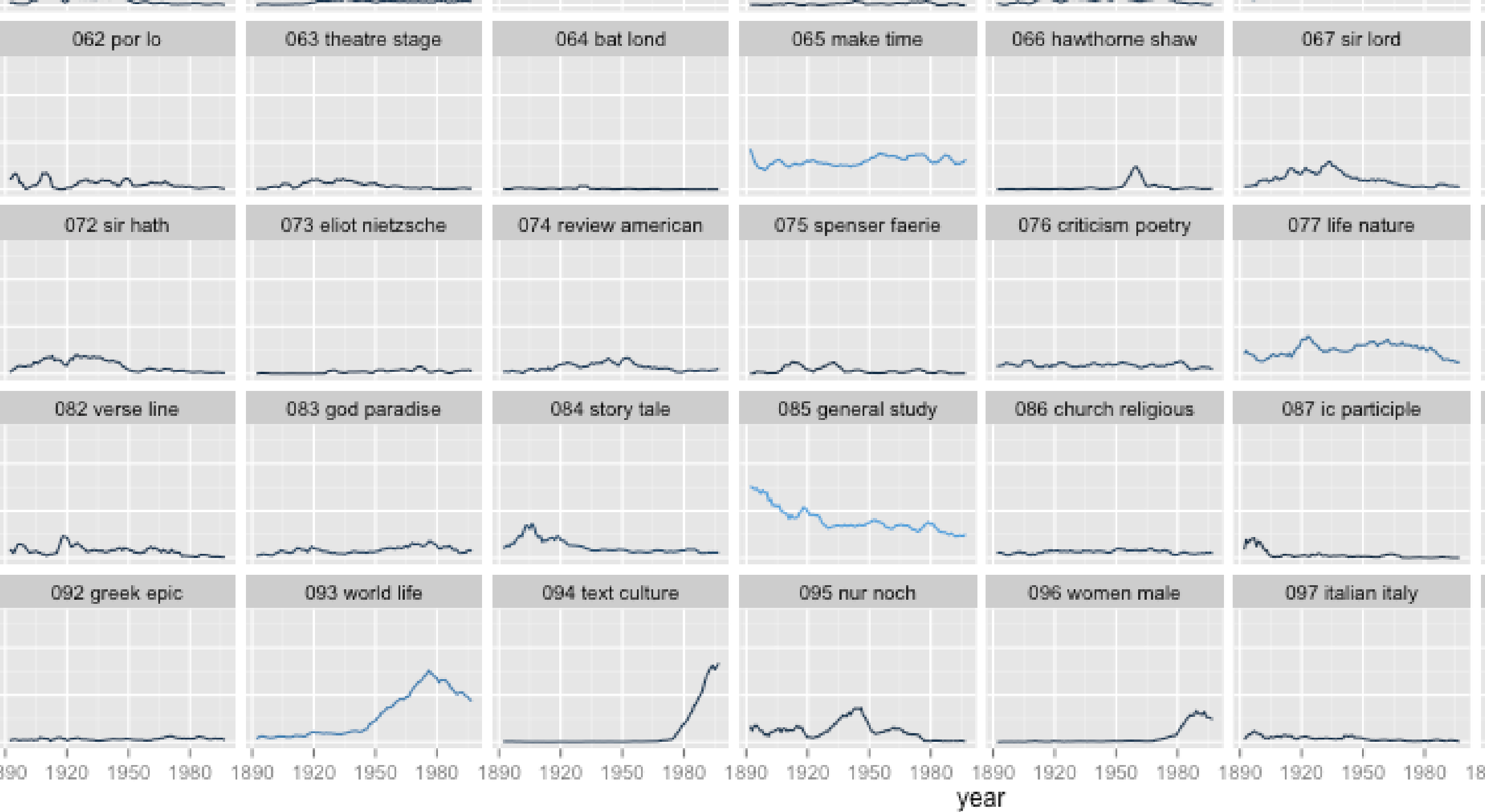
What Can Topic Models of PMLA Teach Us About the History of Literary Scholarship?

TED UNDERWOOD AND ANDREW GOLDSTONE

Of all our literary-historical narratives it is the history of criticism itself that seems most wedded to a stodgy history-of-ideas approach — narrating change through a succession of stars or contending schools. While scholars like John Guillory and Gerald Graff have produced subtler models of disciplinary history, we could still do more to complicate the narratives that organize our discipline's understanding of itself.



(Underwood
and
Goldstone,
2012)



What's in a Topic Model?

- The concept of topic (or thema) in **functionalist linguistics**?
 - The notion of isotopy in **structuralism and semiotics**?
- The concepts of theme and motif in **thematic criticism**?
 - The **Foucaultian** notion of «discourse»?

“La discussione sulle possibili interpretazioni semiotico-letterarie della nozione di topic model e la constatazione della difficoltà teoriche che esse presentano ci porta ad affermare che in effetti **non è possibile trovare un unico e soddisfacente correlato teorico-letterario** dei risultati di questi metodi di analisi quantitativa”

(Ciotti, 2017)